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### Rotating electric machine, force feedback mini-joystick and related aircraft

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#### Abstract

A rotating electric machine includes a rotor and a wound stator. The wound stator has a stator shaft, wherein the stator coils are inserted in the rotor, and the stator shaft is inserted in the center of the rotor. At least one angular position sensor of the rotor is inserted in the center of the space delimited by the stator shaft and the rotor.

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**Inventors:** Lapôte; Blaise (Moissy-Cramayel, FR), Argenton; Julien (Moissy-Cramayel, FR), Attrazic; Yannick (Moissy-Cramayel, FR), Ferraud; Xavier Marcel Adolphe (Moissy-Cramayel, FR)

**Applicant:** SAFRAN ELECTRONICS & DEFENSE (Paris, FR)

**Family ID:** 78827523

**Assignee:** SAFRAN ELECTRONICS & DEFENSE (Paris, FR)

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*Primary Examiner:* Patel; Tulsidas C

*Assistant Examiner:* Qureshi; Mohammed Ahmed

*Attorney, Agent or Firm:* CHRISTENSEN O'CONNOR JOHNSON KINDNESS PLLC

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## Background/Summary

## CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to FR 2110628, filed Oct. 7, 2021, the disclosure of which is hereby expressly incorporated by reference herein in its entirety.

## TECHNICAL FIELD

(2) The disclosure relates to rotating electric machines, more particularly to rotating electric machines installed in force feedback mini joysticks and to aircraft equipped with such mini joysticks.

## BACKGROUND

(3) Typically, a force feedback mini joystick comprises an electric motor with sensors for the angular position of a rotor of the motor to generate a force path, so that the pilot feels the force feedback through the mini-joystick.

(4) Since this functionality requires a high level of safety, the force path must be doubled to achieve force feedback redundancy.

(5) Achieving force feedback redundancy requires using at least double the position sensors of the electric motor rotor.

(6) The position sensors typically include resolvers.

(7) FIG. 1 shows a longitudinal section of an electric motor 1 comprising a stator 2, a rotor 3 inserted into the stator 2 and four resolvers 4 forming two redundant position sensors of a rotor shaft 5 of the rotor 3.

(8) Each resolver 4 comprises a set of coils 6 including a field coil, a sine coil and a cosine coil arranged on the rotor shaft 5, and an elliptical magnetic flux guide 7 arranged on the stator 2, separated from the set of coils 6 by an air-gap.

(9) The resolvers 4 are stacked on top of each other. This stacking arrangement increases the length of the stator 2 and the rotor shaft 5, so that the electric motor 1 takes up a large amount of space, thus creating the need to provide a large space in a confined area such as an aircraft cockpit.

(10) Furthermore, as the motor 1 is bulky, when fitted in the aircraft cockpit, the risk of collision of the motor 1 with other components present in the cockpit is higher.

(11) Additionally, since the resolvers 4 are stacked on top of each other, the length of the rotor shaft 5 is so great that the inertial mass of the rotor decreases the dynamics of the motor 1 and increases the bending moment and the torsional moment of the rotor shaft 5.

## SUMMARY

(12) This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This summary is not intended to identify key features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

(13) In view of the above, the disclosure relates to a rotating electric machine comprising a rotor and a wound stator including a stator shaft, the stator coils being inserted in the rotor and the stator shaft being inserted in the center of the rotor.

(14) At least one angular position sensor of the rotor is inserted in the center of the space delimited by the stator shaft and the rotor.

(15) The insertion of the angular position sensors in the center of the space delimited by the stator coils serves to reduce the size of the rotating electric machine in order to facilitate the integration and mounting of the machine in the aircraft cockpit while reducing risks of collision with other components of the cockpit.

(16) Additionally, as the angular position sensors are housed in the active part of the rotating electric machine, the angular position sensors are protected from shocks and magnetic disturbances by the rotor of the rotating electric machine.

(17) Furthermore, reducing the length of the rotor serves to improve the dynamics of the rotor by reducing its inertial mass, reducing the bending moment and the torsional moment of the rotor.

- (18) Preferably, the stator comprises a toroid including the stator coils wound around the toroid.
  - (19) Winding the stator coils around the toroid reduces stator magnetic flux losses, thereby increasing the electromotive force.
  - (20) Advantageously, the toroid comprises two half-toroids, the coils of each half-toroid being powered by a different power system.
  - (21) Should one of the power systems fail, the rotating electric machine powered by the other power system continues to operate.
  - (22) Preferably, the angular position sensor comprises a variable reluctance resolver comprising sets of control coils each including a field coil, a sine coil and a cosine coil arranged on the stator shaft and an elliptical magnetic flux guide arranged on the rotor, the control coils and the magnetic flux guide being separated by an air-gap.
  - (23) Advantageously, the angular position sensor comprises an elliptical permanent magnet arranged on the rotor, and means for measuring the variation of a magnetic field in two perpendicular directions arranged on the stator shaft, the magnet and the measuring means being separated by an air-gap.
  - (24) Preferably, the measuring means include at least one Hall effect sensor.
  - (25) Advantageously, the rotating electric machine includes at least two redundant stacked angular position sensors.
  - (26) Preferably, a first angular position sensor includes a variable reluctance resolver as defined above, and the second angular position sensor includes an angular position sensor as earlier described.
  - (27) The asymmetric sensor technology enhances the control reliability of the rotating electric machine.
  - (28) Also proposed is a force feedback mini-joystick for aircraft, comprising a rotating electric machine as earlier described.
  - (29) Also proposed is an aircraft, a force feedback mini joystick as earlier described.
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## Description

### DESCRIPTION OF THE DRAWINGS

- (1) The foregoing aspects and many of the attendant advantages of this invention will become more readily appreciated as the same become better understood by reference to the following detailed description, when taken in conjunction with the accompanying drawings, wherein:
- (2) FIG. 1 already mentioned above, illustrates a longitudinal section of an electric motor according to the prior art;
- (3) FIG. 2 schematically illustrates an aircraft according to the disclosure;
- (4) FIGS. 3 and 4 illustrate longitudinal sections of an example of a rotating electric machine according to the disclosure;
- (5) FIG. 5 illustrates a first embodiment of an angular sensor according to the disclosure,
- (6) FIG. 6 illustrates a second embodiment of an angular sensor according to the disclosure, and
- (7) FIG. 7 illustrates an example of a toroid according to the disclosure.

### DETAILED DESCRIPTION

- (8) While illustrative embodiments have been illustrated and described, it will be appreciated that various changes can be made therein without departing from the spirit and scope of the invention.
- (9) Reference is made to FIG. 2 which schematically illustrates an aircraft **10** including a cockpit comprising a force feedback mini joystick **11**.
- (10) The aircraft **10** may comprise an airplane or a helicopter.
- (11) The mini-joystick includes a rotating electric machine **12** that generates a force path, so that the pilot feels the force feedback through the mini-joystick **11**.

- (12) The rotating electric machine **12** may comprise several phases.
- (13) FIGS. **3** and **4** illustrate longitudinal sections of an example of rotating electric machine **12**.
- (14) The rotating electric machine **12** comprises a rotation axis (A), a rotor **13** rotating around the axis (A) and a stator **14**.
- (15) The rotor **13** comprises two rows of permanent magnets **15**, **16** located on two parallel inner surfaces of the rotor **13**, so as to form a cylindrical cavity **17** delimited by the two rows of permanent magnets **15**, **16**.
- (16) The rotor **13** is recessed at its center.
- (17) The stator **14** comprises a stator shaft **18** inserted into the recessed center of the rotor **13**.
- (18) The rotor **13** is rotatably held in the stator **14** using ball bearings **19**, for example, mounted on the stator shaft **18** and the rotor **13**.
- (19) A first ball bearing **19** is arranged at a first end of the cylindrical cavity **17**, and a second ball bearing **19** is arranged at a second end of the cylindrical cavity **17**.
- (20) The stator **14** further comprises a toroid **20** including stator coils **21** wound around the toroid **20**, and typically made of copper.
- (21) The toroid **20** is inserted into the cylindrical cavity **17** of the rotor **13**, so that when the stator coils are energized, the rotor **13** is rotated about the axis of rotation (A).
- (22) As the stator coils **21** are wound around the toroid **20**, the entire surface of the toroid **20** generates a magnetic flux contributing to the generation of a mechanical torque by the machine **12**.
- (23) The stator magnetic flux losses are reduced, thereby increasing the electromotive force. By so doing, the power and torque developed by the machine **12** are increased so that the power density of the machine **12** is optimal.
- (24) Additionally, the generated torque does not display torque ripple or detent torque.
- (25) Alternatively, the stator coils **21** are wound on teeth of the toroid **20** arranged opposite one of the two rows of permanent magnets **15**, **16**.
- (26) The rotating electric machine **12** also comprises sensors for the angular position of the rotor **13** including four resolvers **22**, **23**, **24**, **25** with variable reluctance, stacked on each other and inserted in the space delimited by the stator shaft **18** and the rotor **13**.
- (27) According to an embodiment, the resolvers **22**, **23**, **24**, **25** are stacked and inserted in the space delimited by the stator coils **21** and between the stator shaft **18** and the rotor **13**.
- (28) Preferably, the angular position sensors are inserted in the centre of the space delimited by the stator coils **21**.
- (29) Inserting the angular position sensors in the center of the space delimited by the stator coils **21** helps to reduce the overall dimensions of the rotating electric machine **12**, making it possible to shorten the length of the rotor **13** along the axis of rotation (A), and the overall dimensions of the rotating electric machine **12**.
- (30) Reducing the overall dimensions of the rotating electric machine **12** makes it easier to incorporate and mount the machine in the aircraft cockpit **10** by reducing risks of collision with other components of the cockpit.
- (31) Additionally, as the angular position sensors are housed in the active part of the rotating electric machine **12**, in the space delimited by the stator coils, the angular position sensors are protected from shocks and magnetic disturbances by the rotor of the rotating electric machine **12**.
- (32) Reducing the length of the rotor **13** serves to improve the dynamics of the rotor **13** by reducing its inertial mass, reducing the bending moment and the torsional moment of the rotor **13**.
- (33) Since the angular position sensors are arranged between the two ball bearings **19** used to rotatably hold the rotor **13** in rotation relative to the stator **14**, the stiffness of the stator shaft **18** and the rotor **13** is increased, thereby allowing more accurate angular position measurements.
- (34) It is assumed that the four resolvers **22**, **23**, **24**, **25** are identical.
- (35) Only the first resolver **22** is detailed.
- (36) FIG. **5** illustrates a view of the first resolver **22**. The first resolver **22** comprises a yoke **26**

arranged on the stator shaft **18** and an elliptical magnetic flux guide **27** arranged on the rotor **13** and surrounding the yoke **26**.

(37) The yoke **26** comprises ten coil heads **28a** to **28j** evenly distributed over the yoke **26**.

(38) Of course, the yoke **26** may comprise a different number of coil heads evenly distributed over the frame **26**.

(39) A set of control coils **29** including a field coil **30**, a sine coil **31** and a cosine coil **32** are wound on each coil head **28a** to **28j**.

(40) The field coil **30**, the sine coil **31** and the cosine coil **32** of a set of control coils **29** are separated by an insulator (not shown) preventing a short circuit between the coils.

(41) The sets of control coils **29** and the magnetic flux guide **27** are separated by an air-gap **33**.

(42) Connection terminals **34**, **35**, **36** are connected to the field coils **30**, the sine coils **31** and the cosine coils **32**.

(43) When the field coil is excited by a sinusoidal signal, signals are generated in the sine and cosine coils.

(44) The characteristics of the signals depend on the position of the magnetic flux guide **27** connected to the rotor **13**. Analyzing the generated signals helps to determine the angular position of the rotor **13** relative to the stator **14** and to control the position of the rotor **13**, in order to generate force feedback in the mini-joystick **11**.

(45) The resolvers **22**, **23**, **24**, **25** are redundantly connected to each other in order to improve control reliability of the rotating electric machine **12** and to facilitate diagnosis should any of the resolvers fail.

(46) For example, a first and a second resolver **22**, **23** are redundantly connected to a first control channel, and a third and a fourth resolver **24**, **25** are redundantly connected to a second control channel so that if the resolvers of the same control channel do not output identical signals, the control channel is considered to have failed.

(47) The failed control channel is deactivated and control signals from the other control channel are taken into consideration.

(48) Of course, when the rotating electric machine **12** is implemented for applications not requiring such a high reliability level, the machine **12** may comprise a single angular position sensor.

(49) According to a second embodiment, the angular position sensor comprises an elliptical permanent magnet **37** arranged on the rotor **13** and means **38** for measuring the variation of a magnetic field along two perpendicular directions arranged on the stator shaft **18** (FIG. 6).

(50) The elliptical permanent magnet and the measuring means are separated by an air-gap **39**.

(51) For example, the measuring means may include at least one Hall effect sensor.

(52) In order to further enhance the control reliability of the rotating electric machine **12**, the angular position sensors comprise resolvers and Hall effect sensors associated with the permanent magnets to obtain an asymmetric sensor technology.

(53) For example, the first and third resolvers **22**, **24** are replaced by Hall effect sensors coupled to the permanent magnets.

(54) FIG. 7 illustrates an example of a toroid **20** comprising the stator coils **21**.

(55) The toroid **20** is composed of two half-toroids **40**, **41** on which the stator coils **21** are wound.

(56) In order to further increase the control reliability of the rotating electric machine **12**, the stator coils **21** of each half-toroid **40**, **41** are powered by a different power system **42**, **43**, for example the three-phase type.

(57) Should any of the power systems **42**, **43** fail, the three-phase type rotating electric machine **12** continues to operate, powered by only the other power system.

(58) Of course, the rotating electric machine **12** is of the polyphase type and may comprise more than three phases.

## Claims

1. A rotating electric machine comprising: a rotor; and a wound stator including a stator shaft and stator coils, the stator coils being inserted in the rotor and the stator shaft being inserted in a center of the rotor, wherein at least one angular position sensor of the rotor is inserted in a center of a space delimited by the stator shaft and the rotor, and in a center of a space delimited by the stator coils.
  2. The electric machine according to claim 1, wherein the stator further includes a toroid, wherein stator coils are wound around the toroid.
  3. The electric machine according to claim 2, wherein the toroid comprises two half-toroids, coils of each half-toroid being powered by a different power system.
  4. The machine according to claim 1, wherein the angular position sensor comprises a variable reluctance resolver comprising sets of control coils each including a field coil, a sine coil and a cosine coil arranged on the stator shaft and an elliptical magnetic flux guide arranged on the rotor, the control coils and the magnetic flux guide being separated by an air-gap.
  5. The machine according to claim 1, wherein the angular position sensor comprises an elliptical permanent magnet arranged on the rotor, and means for measuring a variation of a magnetic field in two perpendicular directions arranged on the stator shaft, the magnet and the measuring means being separated by an air-gap.
  6. The machine according to claim 5, wherein the measuring means include at least one Hall effect sensor.
  7. The machine according to claim 1, further comprising at least two redundant stacked angular position sensors.
  8. The machine according to claim 7, wherein a first angular position sensor includes a variable reluctance resolver comprising sets of control coils each including a field coil, a sine coil and a cosine coil arranged on the stator shaft and an elliptical magnetic flux guide arranged on the rotor, the control coils and the magnetic flux guide being separated by an air-gap, and the second angular position sensor comprises an elliptical permanent magnet arranged on the rotor, and means for measuring a variation of a magnetic field in two perpendicular directions arranged on the stator shaft, the magnet and the measuring means being separated by an air-gap.
  9. A force feedback mini-joystick for aircraft comprising a rotating electric machine according to claim 1.
  10. An aircraft including a force feedback mini-joystick according to claim 9.
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